Quantitative Methods for Business and Economics

Lecture 6 - Limits

Daniel Sánchez Pazmiño

2026

Why limits?

- Last lecture, we defined the derivative as:

derivative

$$\boxed{f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}} \approx \frac{x_2 - x_1}{x_1}$$

- We used the word "limit" without really explaining what it means. Let's fix that.
- A limit just describes the value a function *gets closer and closer to*, as its input gets closer and closer to some point.

A numerical example

$$\frac{1^2-1}{1-1} = \frac{0}{0}$$

- Consider $\boxed{f(x) = \frac{x^2-1}{x-1}}$. Plugging in $x = 1$ gives $\frac{0}{0}$ — undefined.
- But let's see what happens as x gets closer to 1:

x	0.9	0.99	0.999	1.001	1.01	1.1
$f(x)$	1.9	1.99	1.999	2.001	2.01	2.1

$\simeq 2$ $\simeq 2$

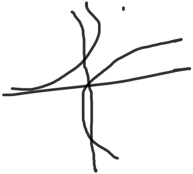
- Even though $f(1)$ doesn't exist, $f(x)$ keeps getting closer to 2. We say:

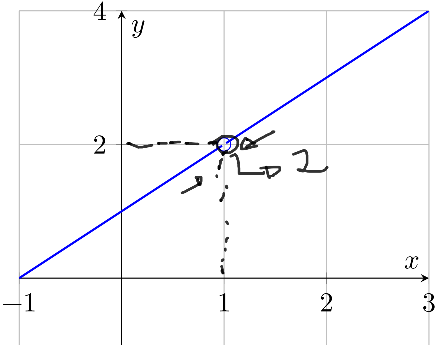
$$\lim_{x \to 1} \frac{x^2-1}{x-1} = 2$$

Graphing it

- $f(x) = \dfrac{x^2 - 1}{x - 1}$ is just the line $y = x + 1$ with a "hole" punched out at $x = 1$.

rational





y = mx + b

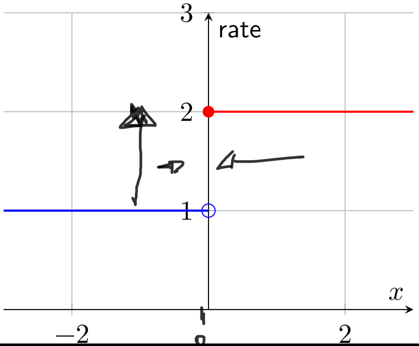
- The open circle is the hole. Even though the function is never actually *at* 2, it gets arbitrarily close — that's the limit.

When there's no single value to approach

- A shipping company charges a flat rate that jumps at a weight threshold:

"fn by parts"

$$f(x) = \begin{cases} \$1 & x < 0 \text{ (light package)} \\ \$2 & x \geq 0 \text{ (heavy package)} \end{cases}$$



The practical rule: just plug in

$$\lim_{x \to 3} (\ldots) =$$

- For almost every function we'll use — polynomials, and fractions where the denominator isn't zero — the limit is simply the value of the function at that point.

$$\lim_{x\to 3}(2x^2 - 5) = 2(3)^2 - 5 = 13$$

- No table, no graph needed. Just substitute and evaluate.
- The only time this shortcut fails is when substitution gives you $\frac{0}{0}$.

When plugging in gives 0/0

- Getting $\frac{0}{0}$ doesn't mean the limit doesn't exist — it means the expression can be simplified first.
- Revisit $\lim_{x\to 1} \dfrac{x^2-1}{x-1}$: substitution gives $\frac{0}{0}$. Factor the numerator and cancel:

$x + 1$

$$\frac{x^2-1}{x-1} = \frac{\cancel{(x-1)}(x+1)}{\cancel{x-1}} = x+1 \quad (x \neq 1)$$

- Now substitution works on the simplified version:

$$\lim_{x\to 1}(x+1) = 2$$

- Matches what we saw in the table earlier.

Try one yourself: an example

- Find $\lim_{x\to 2} \dfrac{x^2-4}{x-2}$. → "difference of squares" $= (x^2 - 4) (x^2+4)$

↳ factorization

- Substitution gives $\frac{0}{0}$, so factor:

$$\frac{x^2-4}{x-2} = \frac{\cancel{(x-2)}(x+2)}{\cancel{x-2}} = x+2$$

$$\lim_{x\to 2} \frac{x^2-4}{x-2} = \lim_{x\to 2}(x+2) = 4$$

(1) Table of values
(2) Drawing chart

(3) Factorizing (simplify)

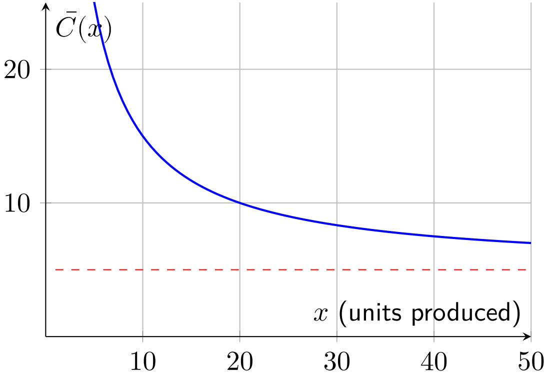
Application: what happens in the long run?

- We can also ask what a function approaches as x grows very large — useful for describing long-run behavior.
- A firm's average cost per unit is $\bar{C}(x) = \dfrac{100}{x} + 5$, where \$100 is fixed cost spread over x units and \$5 is the variable cost per unit.
- As x grows, the fixed-cost share shrinks toward zero, so:

$$\lim_{x \to \infty} \left(\frac{100}{x} + 5 \right) = 5$$

- In words: at very high volumes, average cost approaches the variable cost of \$5 — but never quite reaches it.

Application: the graph



- The dashed line is a **horizontal asymptote**: the value the curve keeps approaching but never crosses.

A faster shortcut: L'Hôpital's Rule

- Factoring works, but isn't always convenient. If substitution gives $\frac{0}{0}$ (or $\frac{\infty}{\infty}$), there's a shortcut:

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$$

- In words: differentiate the top and the bottom **separately** (this is *not* the quotient rule), then substitute again.
- Example: $\lim_{x \to 0} \dfrac{e^x - 1}{x}$ gives $\frac{0}{0}$. Differentiate top and bottom:

$$\lim_{x \to 0} \frac{e^x - 1}{x} = \lim_{x \to 0} \frac{e^x}{1} = e^0 = 1$$

Continuity, in plain terms

- A function is **continuous** at a point if you can draw its graph through that point without lifting your pen.
- Equivalently: the limit as you approach the point matches the actual value of the function there.
- This is exactly why "just plug in" works for polynomials, exponentials, and logs — they're continuous everywhere they're defined.
- Holes (like our first example) and jumps (like the shipping rate) are the two ways continuity breaks down — and exactly when we need the extra tools from this lecture.

Back to derivatives

- Recall the derivative definition from last lecture:

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

- Plugging in $h = 0$ directly always gives $\frac{0}{0}$ — it's indeterminate by construction.
- That's exactly why we rely on the differentiation *rules* (power, product, chain, etc.) instead of evaluating this limit from scratch every time — they're built-in shortcuts around this problem.

For you

1. Evaluate $\lim_{x \to 4} \dfrac{x^2 - 16}{x - 4}$ by factoring and cancelling.
2. Evaluate the same limit using L'Hôpital's Rule, and check your answers match.
3. A firm's average cost is $\bar{C}(x) = \dfrac{500}{x} + 8$. What does average cost approach as $x \to \infty$?

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- We used the word "limit" without really explaining what it means. Let's fix that.
- A limit just describes the value a function *gets closer and closer to*, as its input gets closer and closer to some point.

Why do business, finance, and economics care?

- **Marginal analysis is a limit.** Marginal cost, marginal revenue, and marginal profit are all derivatives — and every derivative is defined as a limit. You cannot rigorously justify "the extra cost of one more unit" without one.
- **Long-run behavior is a limit.** Does average cost ever truly hit a floor? Does a market ever fully saturate? These "as x gets huge" questions are exactly the *limits at infinity* we'll cover today.
- **Continuous growth is a limit.** Back in the logarithms lecture, e itself was defined as a limit ($e = \lim_{n\to\infty}(1 + 1/n)^n$) — continuous compounding, population growth, and depreciation models all lean on this.
- In short: limits aren't abstract math for its own sake — they're the rigorous foundation underneath tools you already use (derivatives) and are about to use (long-run/asymptotic reasoning).

A numerical example

- Consider $f(x) = \dfrac{x^2 - 1}{x - 1}$. Plugging in $x = 1$ gives $\frac{0}{0}$ — undefined.
- But let's see what happens as x gets closer to 1:

x	0.9	0.99	0.999	1.001	1.01	1.1
$f(x)$	1.9	1.99	1.999	2.001	2.01	2.1

- Even though $f(1)$ doesn't exist, $f(x)$ keeps getting closer to 2. We say:

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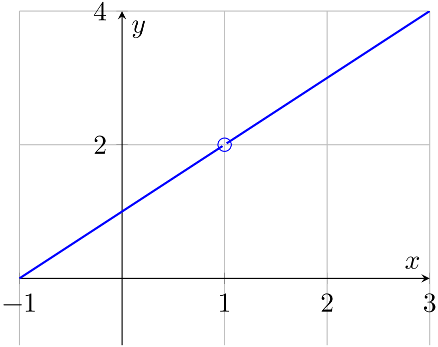

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When there's no single value to approach

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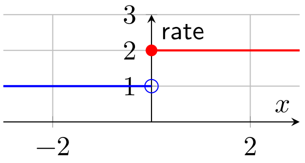

- Approaching from the left gives 1, from the right gives 2 — since these disagree, **the limit at $x = 0$ does not exist**.

The practical rule: just plug in

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$$\lim_{x \to 3}(2x^2 - 5) = 2(3)^2 - 5 = 13$$

- No table, no graph needed. Just substitute and evaluate.
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Limits at infinity: the idea

- So far we've asked "what happens as x approaches a *number*?" Now we ask: what happens as x approaches **infinity** — that is, as x just keeps growing, without bound?

$$\lim_{x \to \infty} f(x)$$

- This is not a number x can actually reach. It's shorthand for: "what value does $f(x)$ keep getting closer to, as x gets larger and larger and larger?"
- This is exactly the tool for "long-run" questions: does average cost ever bottom out? Does a diffusion/adoption curve ever fully saturate?

Limits at infinity: rule 1 — shrinking terms

- The building block you need constantly: **any term of the form $\frac{k}{x^n}$ (with $n > 0$) goes to zero as $x \to \infty$.**

$$\lim_{x\to\infty} \frac{k}{x^n} = 0 \qquad \text{for any constant } k \text{ and } n > 0$$

- Why: as x grows, you're dividing a fixed number k by something enormous — the result shrinks toward zero.
- Examples: $\lim_{x\to\infty} \frac{7}{x} = 0$, $\quad \lim_{x\to\infty} \frac{100}{x^2} = 0$, $\quad \lim_{x\to\infty} \frac{-3}{x} = 0$.
- This single rule is enough to handle a **constant plus a shrinking term**: $\lim_{x\to\infty} \left(5 + \frac{100}{x}\right) = 5 + 0 = 5$.

Limits at infinity: rule 2 — ratios of polynomials

- For $\lim_{x\to\infty} \frac{\text{polynomial}}{\text{polynomial}}$, compare the **highest power (degree)** of x in the numerator vs. the denominator. There are exactly three cases:

1. **Same degree** $\rightarrow$ the limit is the ratio of the **leading coefficients** (the numbers in front of the highest powers).
2. **Numerator's degree is smaller** $\rightarrow$ the limit is 0.
3. **Numerator's degree is larger** $\rightarrow$ there is no finite limit — the function grows without bound.

- The intuition behind all three: for huge x, only the *highest* power in each polynomial matters — every lower-power term becomes comparatively negligible.

Worked example: case 1 (same degree)

- Find $\lim_{x\to\infty} \dfrac{3x^2+5x}{2x^2-1}$. Both top and bottom have degree 2.
- Divide every term by x^2 (the highest power present) to make the shrinking terms explicit:

$$\frac{3x^2+5x}{2x^2-1} = \frac{3+5/x}{2-1/x^2}$$

- Now apply rule 1: $5/x \to 0$ and $1/x^2 \to 0$ as $x \to \infty$:

$$\lim_{x\to\infty} \frac{3x^2+5x}{2x^2-1} = \frac{3+0}{2-0} = \frac{3}{2}$$

- Shortcut check: ratio of leading coefficients is $3/2$ — matches.

Worked example: case 2 (smaller numerator degree)

- Find $\lim_{x\to\infty} \dfrac{4x+1}{x^2+3}$. Numerator has degree 1, denominator has degree 2 — numerator is smaller.
- Divide every term by x^2 (the highest power in the *denominator*):

$$\frac{4x+1}{x^2+3} = \frac{4/x + 1/x^2}{1 + 3/x^2}$$

- Every term in the numerator shrinks to 0 by rule 1; the denominator settles at 1:

$$\lim_{x\to\infty} \frac{4x+1}{x^2+3} = \frac{0+0}{1+0} = 0$$

- Makes sense: the denominator grows much faster than the numerator, so the fraction gets squeezed toward zero.

Worked example: case 3 (larger numerator degree)

- Find $\lim_{x\to\infty} \dfrac{x^2+1}{x+3}$. Numerator has degree 2, denominator has degree 1 — numerator is larger.
- Here the numerator grows faster than the denominator, so the fraction keeps growing rather than settling anywhere:

$$\lim_{x\to\infty} \frac{x^2+1}{x+3} = \infty \quad \text{(no finite limit)}$$

- Business reading: if x were, say, total cost over a shrinking denominator, this would signal cost growing *without bound* relative to that denominator — a red flag, not a stabilizing ratio.

Application: what happens in the long run?

- Now the applied version: a firm's average cost per unit is $\bar{C}(x) = \dfrac{100}{x} + 5$, where \$100 is fixed cost spread over x units and \$5 is the variable cost per unit.
- This is exactly rule 1's "constant plus shrinking term" pattern. As x grows, the fixed-cost share shrinks toward zero, so:

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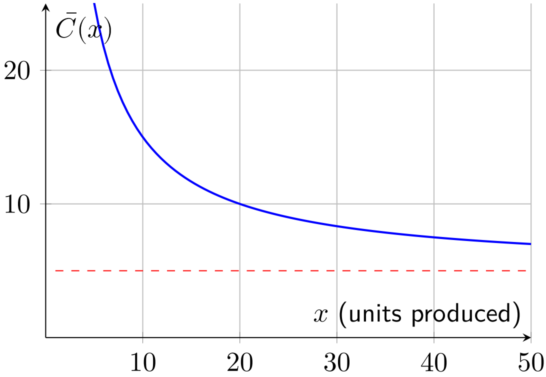
- In words: at very high volumes, average cost approaches the variable cost of \$5 — but never quite reaches it.

Application: the graph



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- That's exactly why we rely on the differentiation *rules* (power, product, chain, etc.) instead of evaluating this limit from scratch every time — they're built-in shortcuts around this problem.

One derivative, two methods: the limit definition

- Let's actually do it from scratch, for $f(x) = x^2$. Start from the definition and substitute:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h}$$

- Expand the numerator:

$$= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} = \lim_{h \to 0} \frac{2xh + h^2}{h}$$

- This is $\frac{0}{0}$ at $h = 0$, but it factors — cancel the common h:

$$= \lim_{h \to 0} (2x + h) = 2x$$

One derivative, two methods: differentiation rules

- Now the fast way, using the power rule from last lecture:

$$f(x) = x^2 \quad \Longrightarrow \quad f'(x) = 2x^{2-1} = 2x$$

- **Same answer, $2x$, both ways.** The power rule isn't a different fact about derivatives — it's a shortcut that already has this limit-cancellation baked in, so you never have to repeat that algebra by hand.
- This is the payoff of the whole lecture: the limit definition is *why* differentiation works; the rules are *how* you actually compute it day to day.

For you

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4. Evaluate $\lim_{x\to\infty} \dfrac{2x+7}{x^2-4}$. (Hint: compare the degrees of the numerator and denominator first.)